Indian Institute of Technology, Bhilai

CSL351: Computer Networks

Assignment - 7 (TCP Congestion Control Algorithm in NS3)

Full Marks: 35 *Deadline: 17/03/2026,11:59PM*

Submission Instructions:

1) Answer all the questions.
2) Deliverables in a .zip:
    a) Submission Guidelines: For each part, create a separate folder. Upload all the folders and the Assignment Report in .zip.
    b) Readable Report [5 Marks for report quality] enumerating steps followed with screenshots for each of the important steps.
        i) Put the screenshots in the report for better clarity
        ii) Your report should **include screenshots of your final results, and you should** explain in the report how the congestion window was traced. Everything in your report needs to be carefully explained.
        iii) **Explain** everything that has been asked in your report and highlight the important parts, and attach each and every code file, script or anything which you used to complete the tasks.
    c) For all the experiments, write the **inferences** that you have observed.
3) Follow the instructions given in the question carefully.
4) You can use LaTeX/Word to make the PDF.
5) The naming of the zip file should strictly follow the given format: <Roll_No>_<Name>_<Assignment No>. If your name is Alex, your roll number is B23CS055, then the filename should be: B23CS055_Alex_07.zip
6) **Any case of plagiarism will be considered an unethical practice, and appropriate action will be taken against the offender.**

Objectives:

1. **Comprehend and Differentiate TCP Variants:**
    a. Explore and understand the workings of different TCP congestion control algorithms, including their characteristics, strengths, and weaknesses.
    b. Compare algorithms from loss-based, delay-based, and hybrid categories, with an emphasis on TCP Cubic and its performance in various scenarios.

2. **Analyze TCP Congestion Dynamics:**
    a. Simulate different TCP variants in NS-3 to study their behavior in response to network congestion.
    b. Understand how congestion window dynamics, such as slow start, congestion avoidance, and cwnd reductions, affect throughput and latency.
3. **Evaluate the Impact of Network Parameters:**
    a. Investigate the effect of varying bandwidth, latency, and MTU size on TCP performance metrics such as throughput, RTT, and congestion window.
    b. Analyze how these parameters influence different congestion control algorithms under similar network conditions.
4. **Visualize, Generate and Interpret Results:**
    a. Generate and interpret graphs for throughput, congestion window, RTT using tools like Matplotlib or Gnuplot.
    b. Create detailed reports with step-by-step explanations, including screenshots of results and code, highlighting key observations and inferences.
    c. Automate network simulation tasks as well so as to save time by just running a single script and letting it take care of the generation of results.

Description:

Try to complete Part 2 during the lab hours and the rest by the mentioned deadline.

PART 1: Study of TCP Congestion Control Algorithms [20 Points]

Explain and compare any four TCP congestion control algorithms (TCP Variants).

Of the four TCP congestion control algorithms, one must be TCP Cubic, and at least one must be from the loss-based, delay-based, or hybrid categories. Definitions of these categories are given below, and some of the TCP variants are shown in the table:

1. Loss-based TCP variants use packet loss as a congestion indication.
2. Delay-based TCP variants consider packet delay rather than packet loss as a sign of network congestion.
3. In the hybrid TCP variant, both packet loss and packet delay are considered as network congestion.

Category	TCP Variants
Loss Based & Loss + Estimation	TCP-Tahoe, TCP-Reno, TCP- New Reno, TCP-SACK, TCP-FACK, , BIC-TCP, TCP Cubic , TCP Hybla TCP, HS-TCP, H-TCP, S-TCP, LP, TCP Libra , TCP Westwood , TCPW CRB, TWPW BR, DOOR,DSACK, TD FR, TCP -FIT
Delay Based	TCP-Vegas , TCP Vegas-A, TCP New Vegas, FAST-TCP, Nice, TCP Real
Hybrid(Loss & Delay) Based & L+D+ Estimation	Compound TCP, , TCP Jersey , TCP Africa, TCP Veno , TCP Illinois', YeAH TCP, TCP Fusion, , TCP-FNC,TCP-ACC

Note: Refer to research papers for different algorithms. The report should have the following points.

- ➔ Algorithm Details **[6 Points]**
- ➔ Suitable for which scenarios and where it fails **[7 Points]**
- ➔ Compare all four variants in the end and write inferences **[7 Points]**

Note for Part 1: Do not copy and paste from the paper; understand the algorithms. This assignment will be evaluated through a student presentation. TAs can randomly select students to present the submitted report for evaluation.

PART 2: Understanding TCP Congestion Window using NS-3 [10 Points]

For each of the given congestion control algorithms, perform the simulations and answer the following questions. (**Demo_2.cc** file is provided)

Instructions: A single script called "**run.sh**" must be created to execute the ns-3 code for the different algorithms. The results should be saved to a results directory, and the files should be named as **newreno_cwnd, newreno_rtt, veno_cwnd, veno_rtt,** etc. The **automation script** needs to be attached, and for plotting the data files, the respective **gnuplot scripts** need to be attached as well.

A. Newreno
B. Highspeed
C. Veno
D. Vegas

1. Plot the cwnd vs time graph, and describe what you observed, like slow start and congestion avoidance, in detail. **[2 Points]**
2. Find the average throughput for each congestion control algorithm using tshark on the pcap files generated, and state which algorithm performed best. **[2 Points]**
3. How many times did the TCP algorithm reduce the cwnd, and why? **[2 Points]**
4. Check the effect of changing the bandwidth and latency of a point-to-point connection and explain its effect on average throughput. **[2 Points]**
5. Explain in short what is the effect of changing the default MTU size. **[1 Point]**
6. Plot the RTT vs time graph and explain your inferences and observations. **[1 Point]**

** **ns-3 version should be >=3.36**
** **Reference File Paths:**

- ns-allinone-3.44/ns-3.44/examples/error-model/simple-error-model.cc
- ns-allinone-3.44/ns-3.44/examples/tutorial/fifth.cc

Check Web sources for more information.